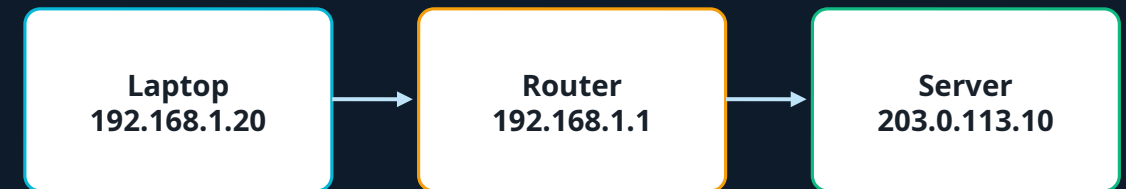


Networking basics

IPv4 • Ethernet • TCP/UDP

A beginner-friendly model for what happens when one program talks to another over a network.

Oleksandr Shcherbakov
Aadesh Baskar



Intro

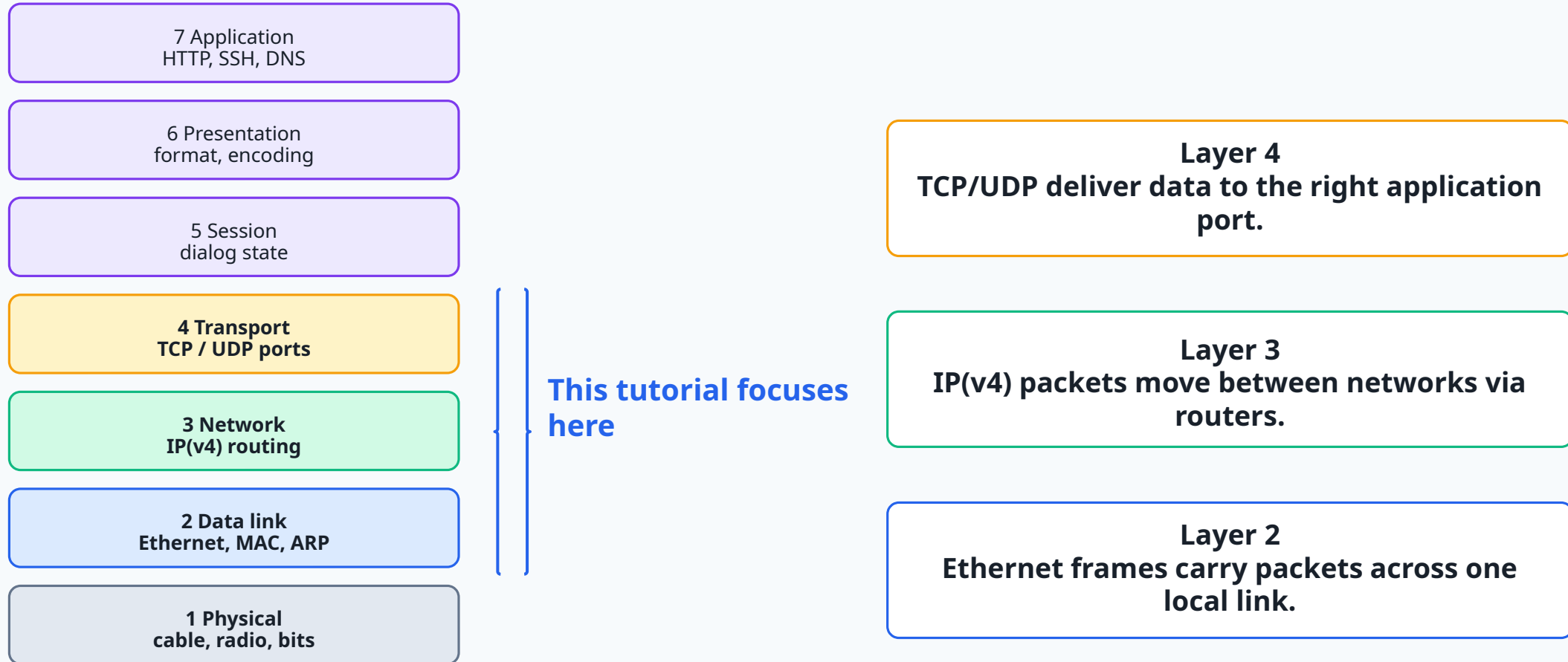
One simple path: app data becomes frames, moves across a LAN, then across IP networks.



- Keep the separation clear: Ethernet is local-link delivery; IP is network-to-network addressing; TCP/UDP is application multiplexing.
- Troubleshooting becomes easier when you ask: which layer is broken?

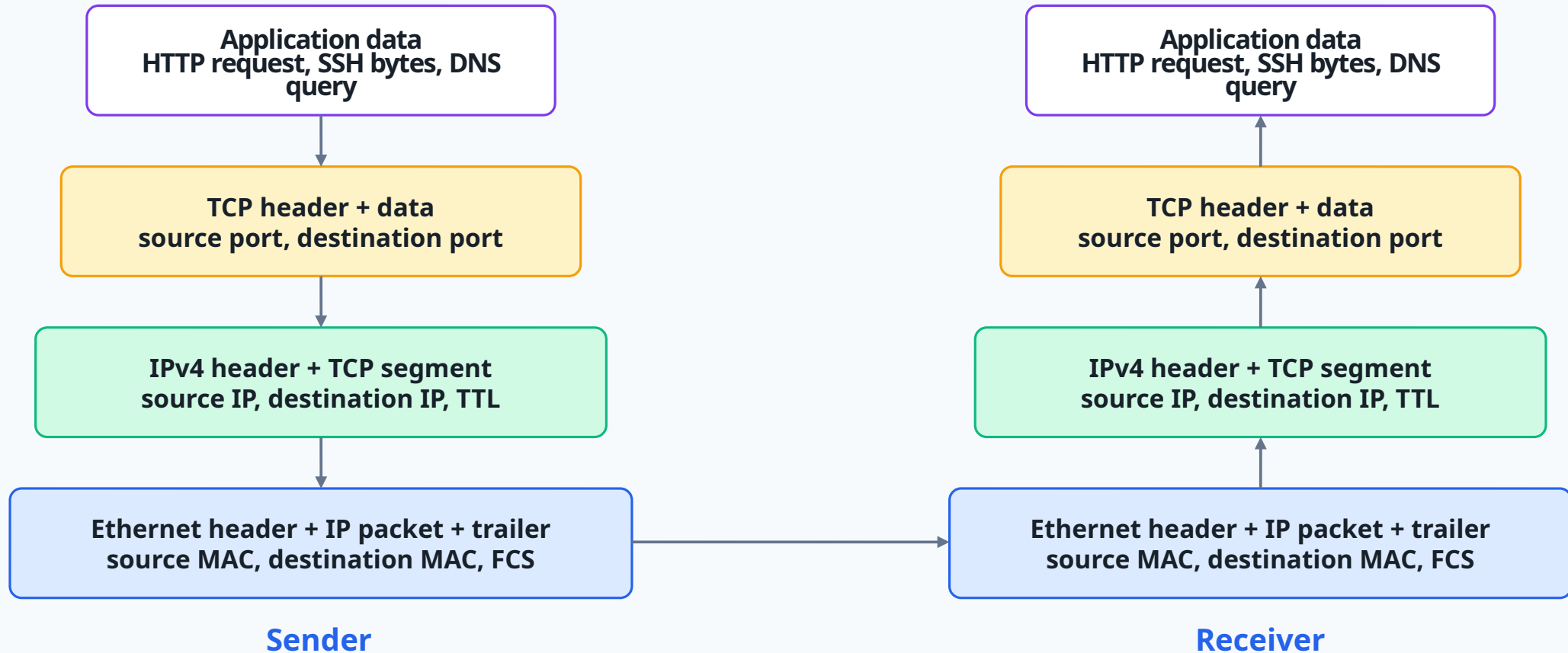
Open Systems Interconnection (OSI) model

Real networks use TCP/IP, but OSI names are useful for explaining responsibilities.



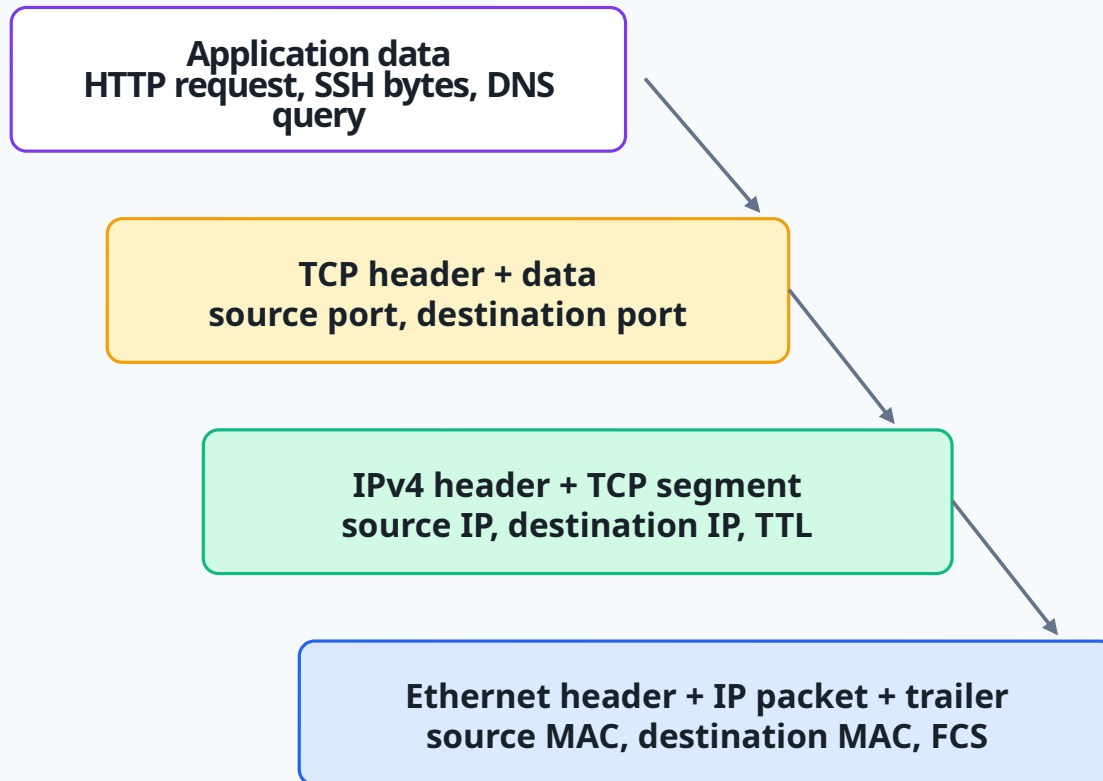
Encapsulation

The sender adds headers on the way down; the receiver removes them on the way up.

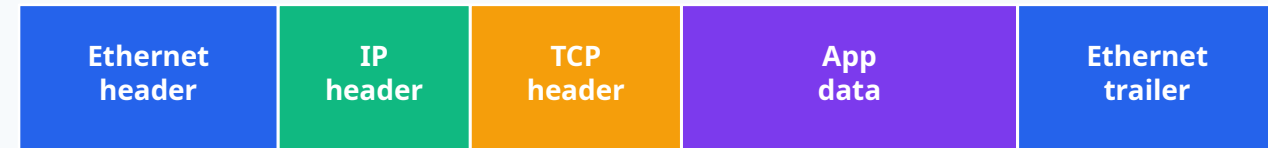


Encapsulation

The sender adds headers on the way down; the receiver removes them on the way up.



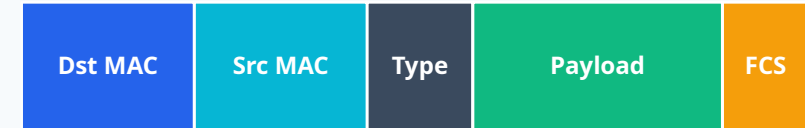
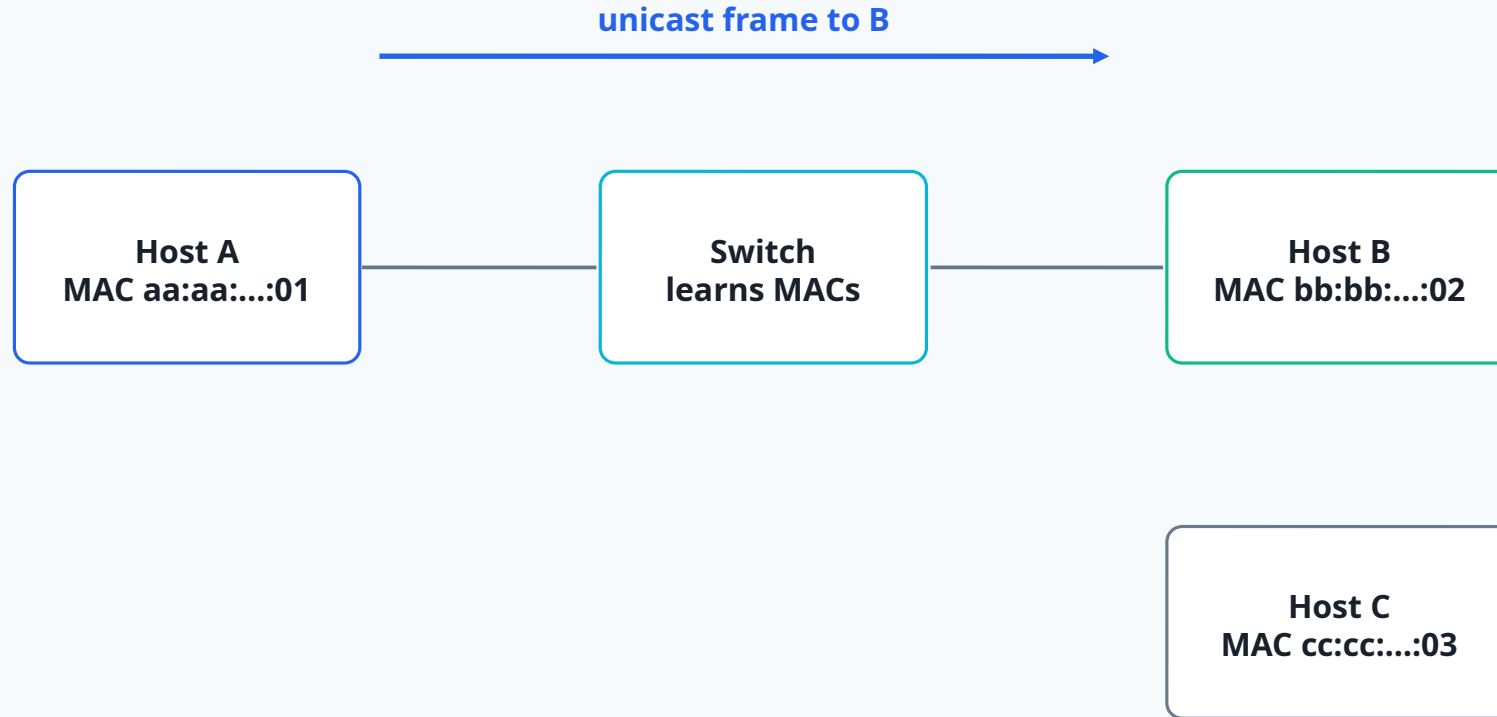
On the wire:



- Headers are control information; payload is what the next layer receives.
- A packet can be re-wrapped into a new Ethernet frame at every router hop.
- Payload size is limited by MTU, so large data is split into smaller pieces.

Ethernet basics: delivery on one local link

Ethernet answers: "which network card on this LAN should receive this frame?"



- A switch forwards frames by destination MAC address.
- Ethernet is not "the Internet"; it is one link or LAN segment.
- The Ethernet payload often contains an IPv4 packet.

MAC addresses: local link identifiers

A MAC address identifies a network interface for Layer 2 delivery.

48 bits
usually written as 6 hex bytes

0A:00:00:00:00:01

0A

00

00

00

00

01

Unicast
one receiver

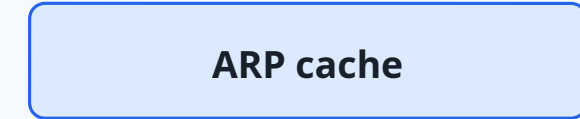
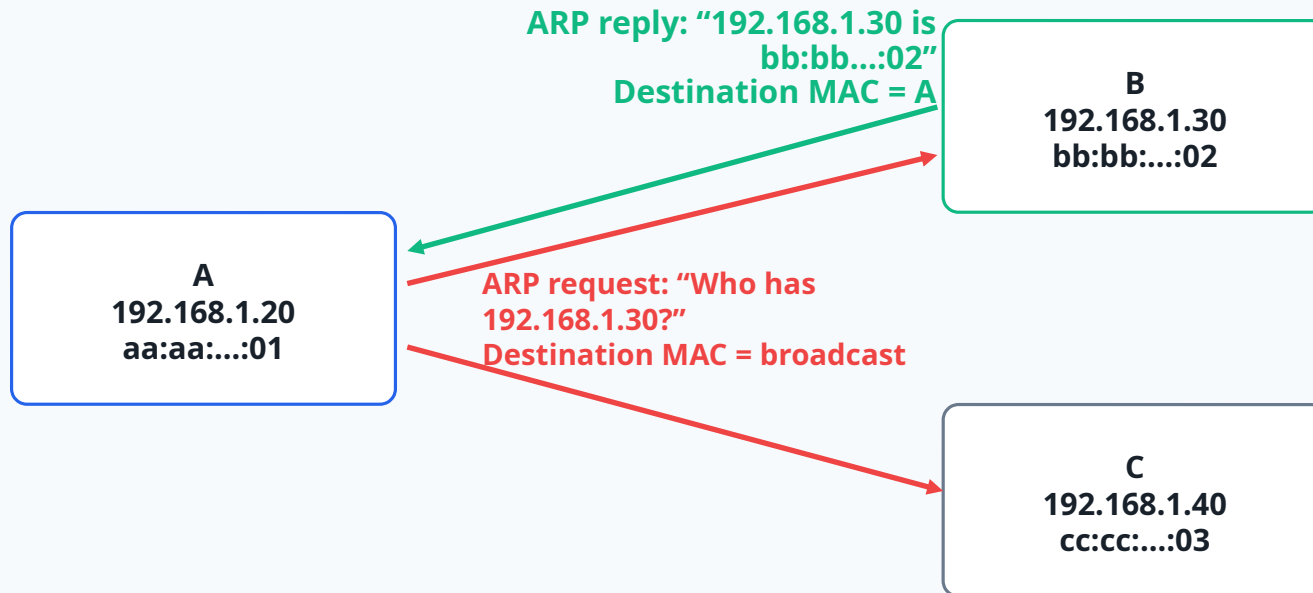
Broadcast
ff:ff:ff:ff:ff:ff

Multicast
one-to-many groups

- MAC addresses can often be changed in software; do not treat them as strong identity.
- When sending to an IPv4 neighbor, the host first needs the neighbor's MAC address.

ARP: finding the MAC address for an IPv4 neighbor

ARP maps an IPv4 address to a MAC address inside the local broadcast domain.



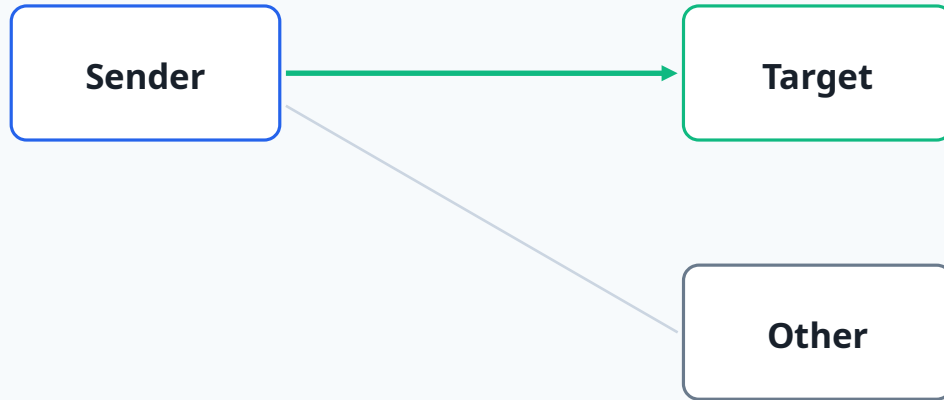
192.168.1.30 → bb:bb:....02
192.168.1.1 → rr:rr:....01

- ARP uses broadcast for the question and unicast for the answer.
- For a remote IP, ARP resolves the gateway's MAC, not the final server's MAC.

Broadcast vs unicast

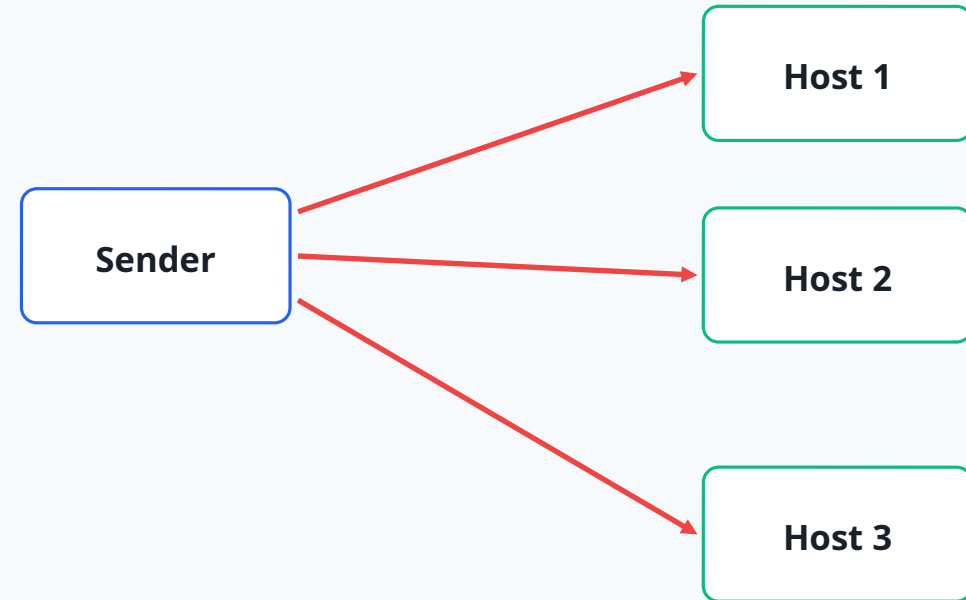
Broadcast is useful but noisy: every host on the LAN must inspect it.

Unicast: one destination MAC



- Common broadcasts: ARP requests, DHCP discovery.

Broadcast: ff:ff:ff:ff:ff:ff

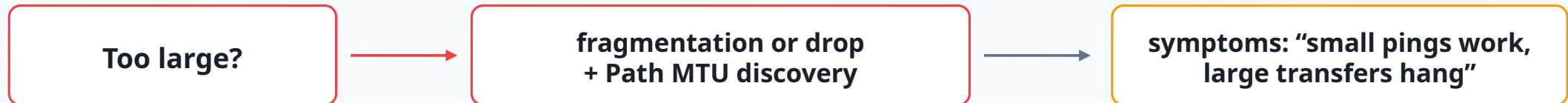
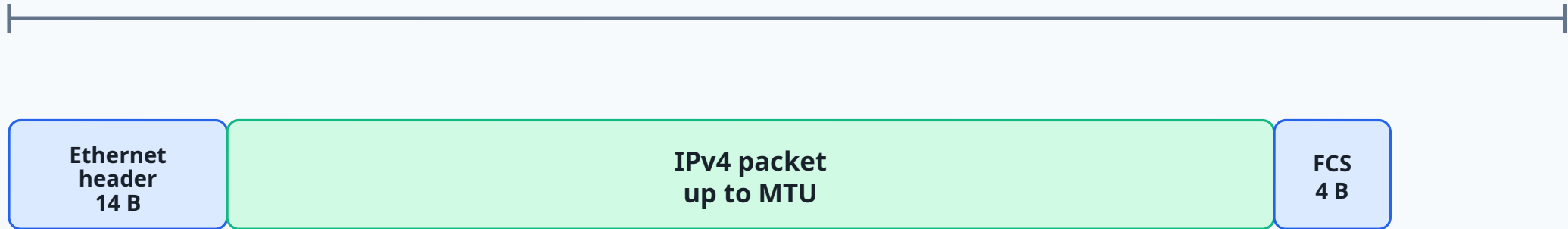


- Routers normally do not forward Layer 2 broadcasts.

MTU: maximum payload size per link

Ethernet's classic MTU is 1500 bytes, but links may differ.

1500 bytes IPv4 payload in a normal Ethernet frame



IPv4 address: a 32-bit number written for humans

An IPv4 address names an interface at Layer 3.

192.168.1.20

192

168

1

20

4 octets × 8 bits = 32 bits

Private ranges often used in
LANs

10.0.0.0/8

172.16.0.0/12

192.168.0.0/16

Special addresses

127.0.0.1 = loopback

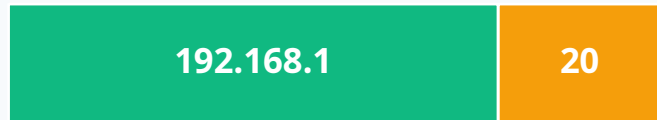
0.0.0.0 = unspecified / all local

- A host can have multiple IP addresses, one per interface or more.
- IP addresses are about network reachability; MAC addresses are about local delivery.

Subnet and mask: which part is the network?

CIDR notation tells how many bits are the network prefix.

192.168.1.20/24



network part

host part

Mask /24
255.255.255.0

Network
192.168.1.0/24

Usable host examples
192.168.1.1
192.168.1.20
192.168.1.254

Broadcast address
192.168.1.255
"All hosts in subnet"

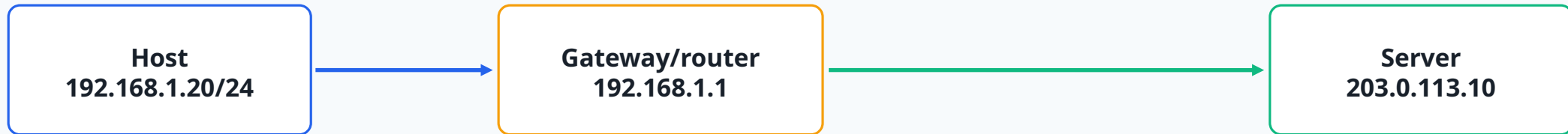
- Same subnet: send directly at Layer 2 after ARP.
- Different subnet: send to the default gateway.

Default gateway: the next hop for other networks

A host sends non-local IPv4 traffic to a router on its own subnet.

Ethernet frame dst MAC = router MAC
IPv4 dst IP = 203.0.113.10

Router unwraps frame, keeps IP packet,
wraps it in a new frame for next link

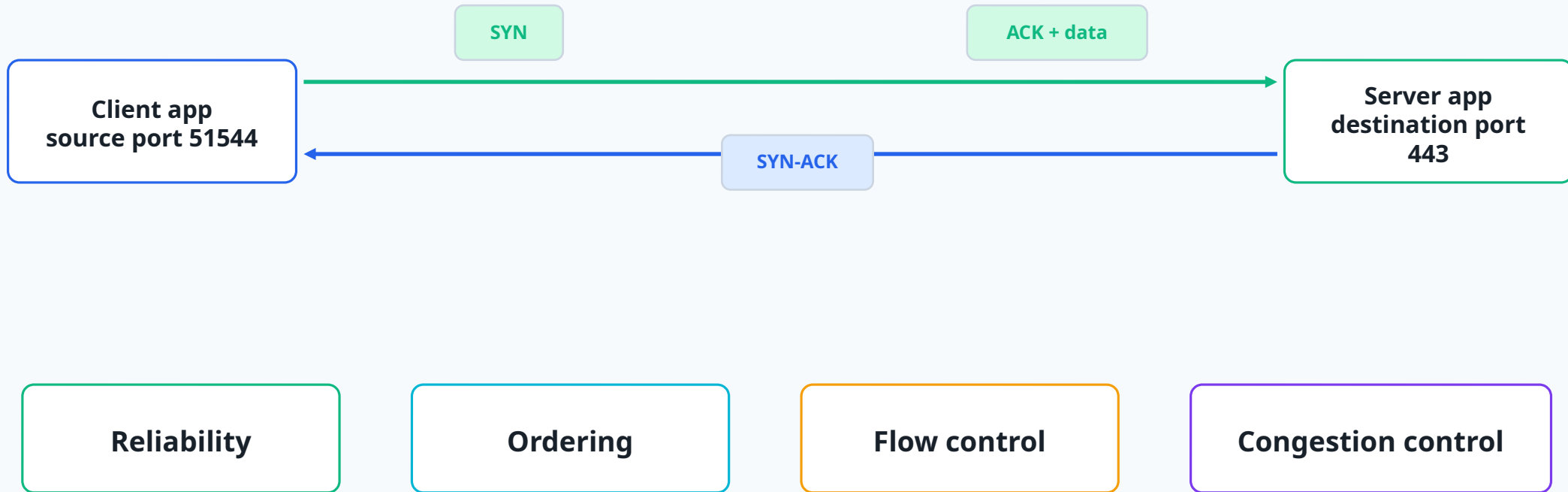


Routing decision on the host

Is destination in my subnet? Yes → ARP for destination. No → ARP for gateway.

TCP: reliable byte stream between two ports

TCP is used when applications need ordered, reliable delivery.



- A TCP connection is identified by: source IP, source port, destination IP, destination port, protocol.

TCP vs UDP: connection vs datagrams

Both use ports; they make different trade-offs.

TCP

- Connection-oriented: handshake before data.
- Reliable and ordered byte stream.
- Used by SSH, HTTPS, many databases.
- Higher overhead, but predictable for apps.

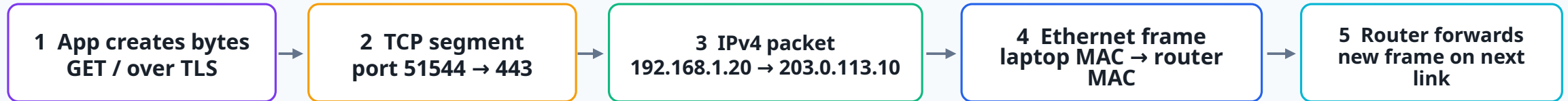
UDP

- No connection: each packet is independent.
- No built-in delivery guarantee or ordering.
- Used by DNS, DHCP, VoIP, games, QUIC.
- Low overhead; applications handle reliability if needed.

Key point: a port number alone is not enough. TCP/443 and UDP/443 are different endpoints.

Putting it together: opening a web page

Follow one HTTPS request from a laptop to a server.



What changes at each router hop?

- Ethernet source/destination MAC addresses change.
- IPv4 source/destination usually stay the same across the path.
- TCP ports stay the same for that connection.

What did the laptop need before sending?

- Its own IP address, subnet mask, and default gateway.
- The gateway MAC address, learned via ARP.
- A destination IP, often learned via DNS.

Cheat sheet: names and where they belong

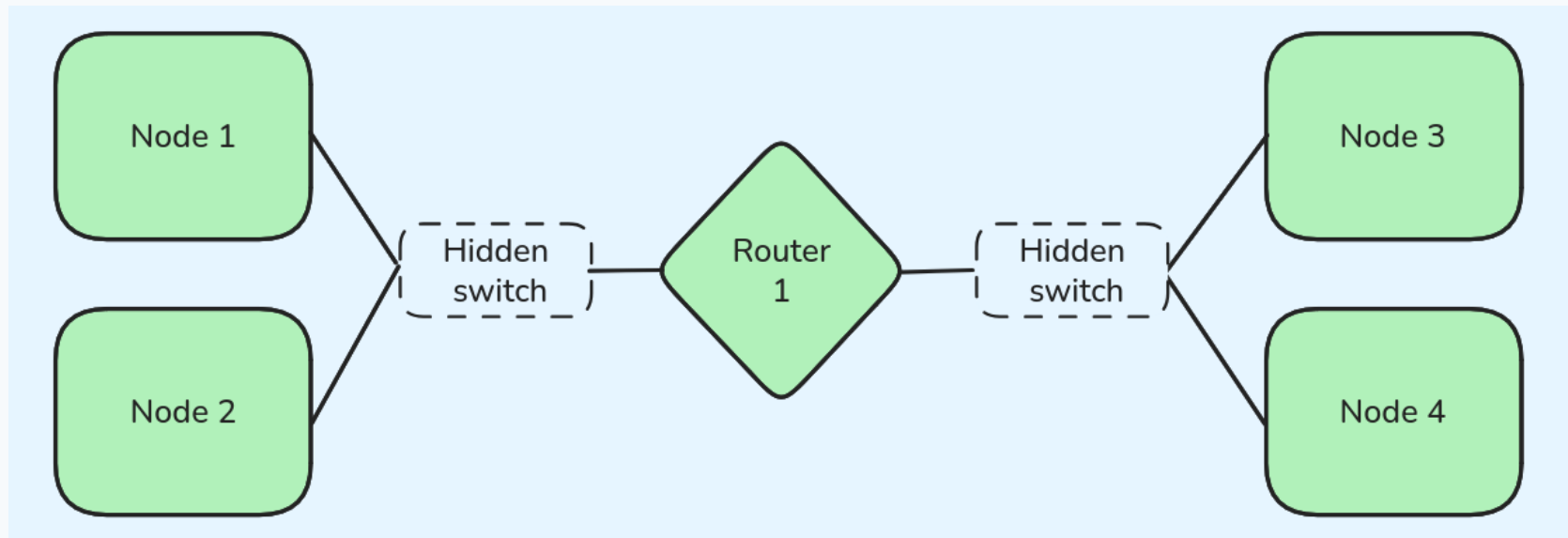
Use this as the final recap.

Term	Layer	Purpose	Example
MAC address	Layer 2	local link delivery	aa:bb:cc:dd:ee:ff
ARP	Layer 2/IPv4 helper	IP → MAC lookup	Who has 192.168.1.1?
MTU	Layer 2/3 boundary	max packet payload on link	1500 bytes common
IPv4 address	Layer 3	network address for interface	192.168.1.20
Subnet/mask	Layer 3	decide local vs remote	/24 = 255.255.255.0
Gateway	Layer 3	router for non-local IPs	192.168.1.1
Port	Layer 4	choose application endpoint	TCP/443, UDP/53

Diagnostic habit: start with the symptom, then ask: link? ARP? IP/subnet/gateway? TCP/UDP port?

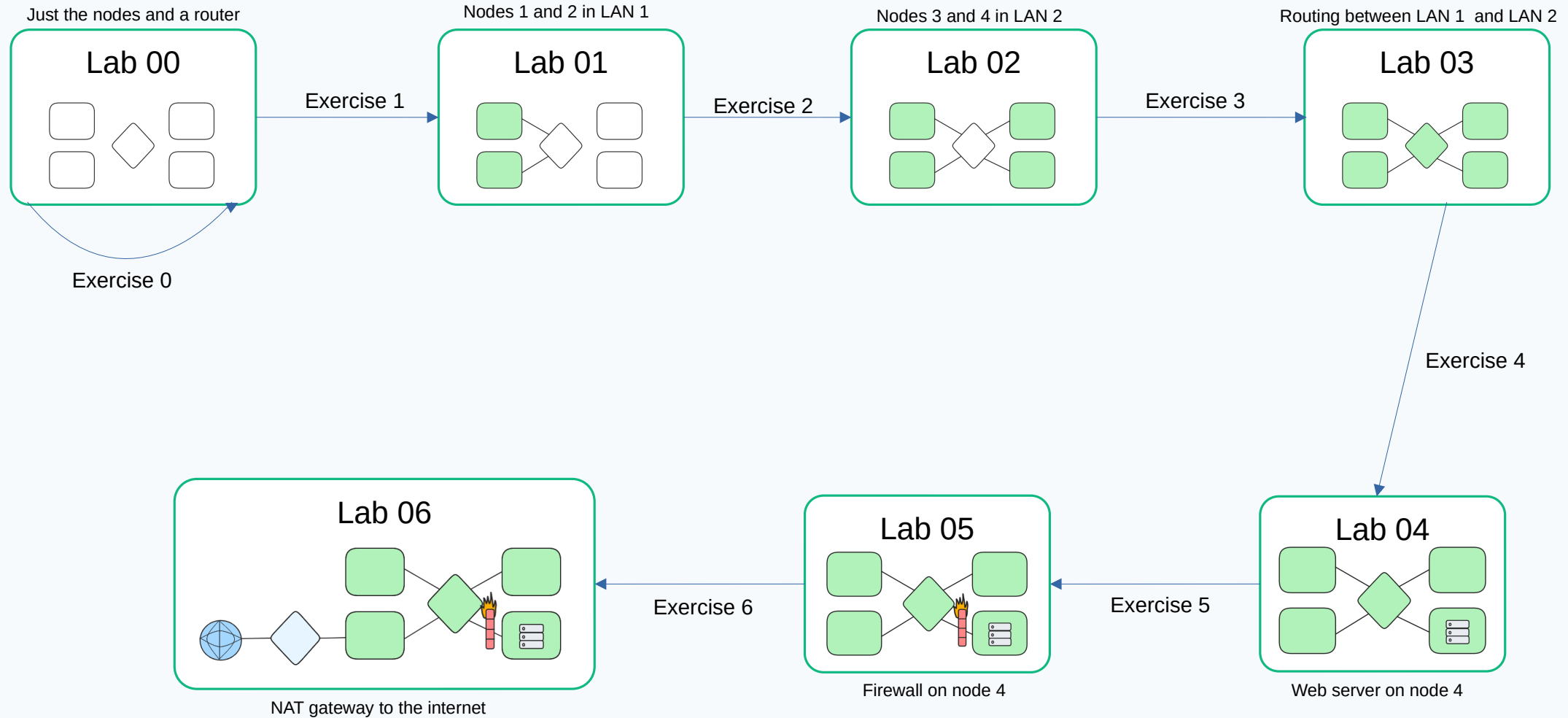
Tutorial setup

- The nodes and routers the participants will interact with are actually containers running in a VM.
- Apache Guacamole can be used to open ssh sessions to the nodes and routers.
- The participants will be able to configure the network interfaces and routes treating their environment as regular linux nodes.
- There are switches connecting the nodes. But, they can be ignored as they are managed by the container engine. For simplicity, assume them to be the equivalent of layer 2 bridges that don't need any configurations.



Tutorial setup

There are 7 exercises. Each lab is a snapshot (checkpoint) of the target state of a particular exercise.



Hands-on

tutorial.403.name